ALiBi's matrix can be approximated by binary block-diagonal matrices, achieving a near-linear time algorithm.

Positional LSH: Binary Block Matrix Approximation for Attention with Linear Biases

Motivation: Long-context attention is computationally expensive, and restricting it to local windows degrades long-range reasoning.

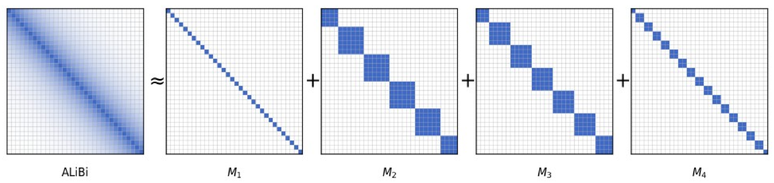
ALiBi (Attention with Linear Biases; Press et al., 2022) adds a linear distance penalty to the attention logits:

$$\langle q_i, k_j \rangle \rightarrow \langle q_i, k_j \rangle - m|i-j|$$

Theoretical Results

1. ALiBi can be approximated by averaging a random sample of binary block-diagonal matrices.
2. The approximation error vanishes as the sample size increases.
3. The computation complexity of the approximated ALiBi attention is near-linear in the context length.

Method: Positional LSH

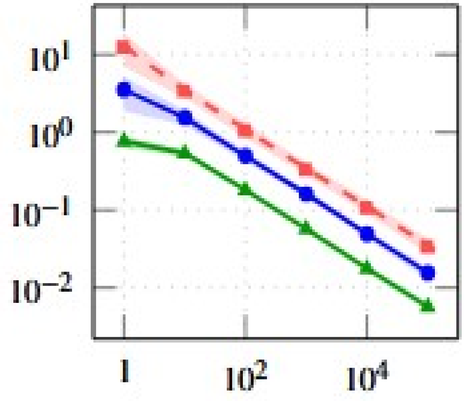


- **LSH (Locality-Sensitive Hashing):** Random hash where near items are more likely to hash together.
- **Positional LSH:** Random attention mask where near positions are more likely to attend each other.
- **LSH for ALiBi:** Random Binning Features (Rahimi & Recht, 2007)

$$e^{-m|i-j|} = \mathbf{Pr}_h[h(i) = h(j)] \approx \frac{1}{s}\sum_t (M_t)_{ij}$$

Empirical Evaluation

- Models: Qwen3-14B, Mistral-7B.
- Matching the theoretical results: The approximation error vanishes as the number of samples increases.



Number of LSH samples

Attention output max norm error

ALiBi Matrix spectral norm error

ALiBi Matrix max norm error

Daniel Wolfson, Tal Wagner
Tel Aviv University